UCR University College Roosevelt

Fall 2026 cohort

Wiskunde en Toepassingen at Utrecht University

Starting interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

Official programme ↗

Mathematical foundations and problem solving
Calculus & Linear Algebra I · 7.5 EC
Proofs in Mathematics · 7.5 EC
Calculus & Linear Algebra II · 7.5 EC
Analysis · 7.5 EC
Probability · 7.5 EC
Introduction to Groups & Rings · 7.5 EC
Introduction to Multivariable Analysis · 7.5 EC
Complex Analysis · 7.5 EC
Differential Equations · 7.5 EC
Statistics · 7.5 EC
Computing, modelling and AI
Programming for Mathematics · 7.5 EC
Methods & Models in Complex Systems · 7.5 EC
Numerical Methods for Physicists & Astronomers · 7.5 EC
Modelling with ODEs & PDEs · 7.5 EC
Introduction to Machine Learning · 7.5 EC
Introduction to Nonlinear Dynamical Systems · 7.5 EC
Introduction to Scientific Computing · 7.5 EC
Climate science, Earth systems and solutions
Atmosphere–Ocean Dynamics · 7.5 EC
Fluid Mechanics & Transport Phenomena · 7.5 EC
Climate Dynamics · 7.5 EC
Geophysical Fluid Dynamics · 7.5 EC
Turbulence in Fluids · 7.5 EC
Research, projects and practical application
Bachelor Thesis · 15 EC

The Utrecht programme includes a more extensive sequence of proof-based mathematics, together with specialist work in climate dynamics and fluid mechanics. At UCR, students can combine quantitative methods with climate science, Earth systems, technology, governance and applied projects. UCR does not offer the same depth or sequence in mathematics and climate science.

Current official example study path within Wiskunde en Toepassingen.

UCR University College Roosevelt

Fall 2026 cohort

Mathematics, Computing and Climate Science

Starting interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

UCR courses ↗

Mathematical foundations and problem solving
Mathematical Methods · 7.5 EC
Logic & Discrete Mathematics · 7.5 EC
Linear Algebra · 7.5 EC
Advanced Calculus · 7.5 EC
Probability & Statistics · 7.5 EC
Linear Systems · 7.5 EC
Computing, modelling and AI
Introduction to Programming · 7.5 EC
Introduction to Data Science · 7.5 EC
Computational & Numerical Methods · 7.5 EC
Machine Learning · 7.5 EC
Algorithms & Data Structures · 7.5 EC
Image Processing & Computer Vision · 7.5 EC
Robotics · 7.5 EC
Optimization Methods in AI · 7.5 EC
Neural Networks & Deep Learning · 7.5 EC
Climate science, Earth systems and solutions
Sustainability Essentials I · 7.5 EC
Sustainability Essentials II · 7.5 EC
Climate Change · 7.5 EC
Dynamic Earth · 7.5 EC
Earth Observation · 7.5 EC
The Science of Evolution · 7.5 EC
Earth System Science Tools · 7.5 EC
Governance, economics and ethics
AI Ethics · 7.5 EC
Academic development
Personal & Professional Development · 7.5 EC

The Utrecht programme includes a more extensive sequence of proof-based mathematics, together with specialist work in climate dynamics and fluid mechanics. At UCR, students can combine quantitative methods with climate science, Earth systems, technology, governance and applied projects. UCR does not offer the same depth or sequence in mathematics and climate science.

UCR University College Roosevelt

Fall 2026 cohort

Climate, Data and Sustainable Technology

Starting interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

UCR courses ↗

Mathematical foundations and problem solving
Mathematical Methods · 7.5 EC
Linear Algebra · 7.5 EC
Probability & Statistics · 7.5 EC
Computing, modelling and AI
Introduction to Programming · 7.5 EC
Introduction to Data Science · 7.5 EC
Computational & Numerical Methods · 7.5 EC
Machine Learning · 7.5 EC
Algorithms & Data Structures · 7.5 EC
Image Processing & Computer Vision · 7.5 EC
Optimization Methods in AI · 7.5 EC
Neural Networks & Deep Learning · 7.5 EC
Climate science, Earth systems and solutions
Sustainability Essentials I · 7.5 EC
Sustainability Essentials II · 7.5 EC
Climate Change · 7.5 EC
Dynamic Earth · 7.5 EC
Earth Observation · 7.5 EC
Earth System Science Tools · 7.5 EC
Renewable System Design · 7.5 EC
Research, projects and practical application
Research in a Sustainable Delta · 7.5 EC
Consumer Product Design · 7.5 EC
Governance, economics and ethics
Ethics · 7.5 EC
Governance of Sustainability · 7.5 EC
Peace & Conflict · 7.5 EC
Academic development
Personal & Professional Development · 7.5 EC

The Utrecht programme includes a more extensive sequence of proof-based mathematics, together with specialist work in climate dynamics and fluid mechanics. At UCR, students can combine quantitative methods with climate science, Earth systems, technology, governance and applied projects. UCR does not offer the same depth or sequence in mathematics and climate science.

UCR University College Roosevelt

Fall 2026 cohort

Environmental Change, Governance and Economics

Starting interests: **I don't really know. I enjoy maths and solving puzzles, but I also care a lot about climate change and would like to do something useful.**

UCR courses ↗

Mathematical foundations and problem solving
Mathematical Methods · 7.5 EC
Probability & Statistics · 7.5 EC
Research Methods & Statistics · 7.5 EC
Computing, modelling and AI
Introduction to Programming · 7.5 EC
Climate science, Earth systems and solutions
Sustainability Essentials I · 7.5 EC
Sustainability Essentials II · 7.5 EC
Climate Change · 7.5 EC
Earth Observation · 7.5 EC
Earth System Science Tools · 7.5 EC
Renewable System Design · 7.5 EC
Life Cycle Analysis & Circular Economy · 7.5 EC
Environmental Biogeochemistry · 7.5 EC
Earth's Resources · 7.5 EC
Research, projects and practical application
Research in a Sustainable Delta · 7.5 EC
Governance, economics and ethics
Ethics · 7.5 EC
Governance of Sustainability · 7.5 EC
Introduction to Economics · 7.5 EC
Law, Society & Justice · 7.5 EC
Foundations of Comparative Politics · 7.5 EC
Business & Economy in the Age of AI · 7.5 EC
Introduction to Public International Law · 7.5 EC
European Union Politics · 7.5 EC
Behavioral Economics · 7.5 EC
Academic development
Personal & Professional Development · 7.5 EC

The Utrecht programme includes a more extensive sequence of proof-based mathematics, together with specialist work in climate dynamics and fluid mechanics. At UCR, students can combine quantitative methods with climate science, Earth systems, technology, governance and applied projects. UCR does not offer the same depth or sequence in mathematics and climate science.